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Personalized learning path generation using large language models and sequential recommendations[☆]

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ABSTRACT

Personalized learning path generation is a crucial challenge in online education, where learners often struggle with information overload and fragmented resources. Existing course recommendation approaches primarily focus on single-step predictions or static learning paths, exempting evolving learning paths. To address this gap, we propose a unified framework that combines sequential recommendation models with large language models (LLMs) to generate dynamic, personalized learning paths. Our approach consists of two stages. First, we develop a sequential recommendation model that captures temporal learning patterns and predicts the next learning resource based on user interactions. This model generates rich embeddings that encode learner preferences and engagement history. In the second stage, we integrate these embeddings into an LLM-based model, enhancing its reasoning capabilities to generate coherent, structured, and context-aware learning paths. By leveraging both sequential behavior and generative AI, our framework improves personalization and adaptivity in learning path recommendations. Extensive experiments on three datasets demonstrate the effectiveness of our approach. Compared to existing sequential models such as GRU4Rec, SASRec, and BERT4Rec, our framework achieves significant improvements in MRR, NDCG, Hit Rate, and Precision (up to 100.51% in MRR, 13% in NDCG, 49.56% in Hit Rate, 155.56% in Precision). Furthermore, our LLM-based model significantly outperforms LLaMA2, achieving relative improvements of 351.85% in ROUGE-1, 1626.67% in ROUGE-2, and 334.48% in ROUGE-L. These results highlight the potential of combining sequential learning with LLMs to enhance personalized learning experiences, offering a novel solution for adaptive and structured knowledge acquisition.

1. Introduction

The rapid expansion of online learning has provided learners with vast educational resources, but it has also introduced challenges in navigating fragmented content. Without proper guidance, learners struggle to identify relevant materials, leading to inefficiencies, reduced motivation, and incomplete knowledge acquisition [1]. Traditional e-learning platforms often lack personalization, failing to account for individual differences in cognitive levels, learning styles, and time constraints. While expert-curated learning paths offer structured guidance, they are labor-intensive and cannot scale to diverse learner needs. As a result, automated and personalized learning path recommendations have emerged as a critical research focus to enhance learning efficiency and engagement [2].

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A learning path generator aims to construct an optimal sequence of knowledge units tailored to a learner's background, goals, and constraints [3]. Unlike conventional recommendation systems that suggest individual courses, learning path generation focuses on sequencing multiple learning resources to provide a structured progression. Personalized path for effective use requires considering factors such as prior knowledge, level of mastery, learning objectives, and availability of time [4]. However, generating adaptive and personalized learning paths remains a complex challenge, particularly when balancing learners' constraints with optimal knowledge sequencing [5].

Recent advancements in language models (LLMs) have introduced new possibilities for educational applications, including personalized feedback, automatic hint generation, and course concept extraction [6–8]. Although LLMs demonstrate promise in processing textual data and generating responses, they often overlook essential behavioral patterns for course recommendations. Additionally, LLM-based recommendation systems can exhibit latent biases inherited from their training data, which are often opaque to both system designers and learners, making it difficult for users to assess the relevance and neutrality of the generated recommendations [9]. Despite these challenges, LLMs provide an opportunity to enhance learning path generation by integrating both semantic and behavioral insights.

Existing LLM-based recommender models, such as CoLLM [10], BIGRec [11], and LLaRA [12], leverage language model embeddings to improve recommendations. However, they primarily focus on next-item prediction rather than the sequential structuring of learning resources. While CoLLM and LLaRA incorporate collaborative user embeddings, they do not explicitly capture temporal learning dependencies essential for pedagogically coherent path construction. This reveals a critical research gap: the lack of a unified framework that integrates sequential behavioral modeling with LLM-based semantic reasoning for personalized learning path generation. From a systems engineering perspective, this limits the ability to design scalable knowledge-based learning systems that align learner behavior, semantic course relationships, and adaptive guidance within a coherent architecture. Inspired by LLaRA [12], this work addresses these gaps by developing a framework that integrates sequential learning patterns with LLM-based recommendation models to generate structured and personalized learning paths. This study aims to develop and evaluate an integrated framework for adaptive learning path generation in MOOCs by combining sequential recommendation modeling with language-model-based learning path synthesis.

We conduct extensive evaluations on three datasets, demonstrating significant performance gains over seven state-of-the-art baselines. Here, performance gains refer to improvements in standard ranking metrics for sequential recommendation (e.g., MRR, NDCG, Hit Rate, Precision) and text-generation quality metrics for learning path generation (e.g., ROUGE scores), as commonly adopted in prior work. Compared to existing sequential recommendation models, our approach *Sequential Learning with User Insights* (SeqInsight) achieves substantial improvements in Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG), Hit Rate, and Precision (up to 100.51% in MRR, 13% in NDCG, 49.56% in Hit Rate, 155.56% in Precision). Furthermore, our LLM-based model, namely *Personalized Learning Path Generation with LLM* (LearnGPT) surpasses the performance of LLaMA 2, achieving relative improvements of 351.85%, 1626.67%, and 334.48% in ROUGE-1, ROUGE-2, and ROUGE-L, respectively. These results highlight the effectiveness of combining sequential embeddings with LLMs for learning path generation.

The rest of the paper is structured as follows. Section 2 reviews related work, providing context for the problem addressed in this study. Section 3 formally defines the learning path generation problem, followed by Section 4, which presents an overview of the proposed solution. Section 5 details the proposed framework for learning path generation, outlining its key components. Section 6 describes the experimental setup, including datasets, evaluation metrics, and baseline models. Section 7 presents the results and findings, demonstrating the effectiveness of the proposed approach. Section 8 discusses various sensitivity analyses related to the proposed model. Finally, Section 9 concludes the paper with key insights, limitations, and directions for future research.

2. Related work

With the rise of online learning platforms and Massive Open Online Courses (MOOCs), the learning path recommendation problem has gained significant attention from researchers and practitioners. This section delineates the various approaches that have been explored to enhance the effectiveness of learning path recommendations. We also discuss the relevant literature on sequential recommendations and LLM-based recommendation literature.

2.1. Learning path generation

The early course recommendation systems relied on collaborative filtering [13] and content-based filtering [14]. These methods identified similarities among users or learning resources, but struggled with data sparsity and cold-start problems. To overcome these challenges, hybrid models combined multiple techniques, such as incorporating similarities in user interest [15,16], but these methods still failed to capture complex and evolving user-course interactions.

Deep learning techniques have significantly advanced the field by capturing intricate behavioral patterns. For instance, deep reinforcement learning [17] and deep belief networks [18] enabled models to better understand student engagement and preferences. Hybrid models have also emerged that integrate association rule mining, collaborative filtering, and content-based filtering [19] to improve recommendation precision. Beyond single-course recommendations, learning path generation focuses on structuring sequences of courses based on users' progress, goals, and engagement. Knowledge graph-based approaches [20] capture complex relationships between learning materials, improving structured learning recommendations. However, these approaches often focus on individual learning trajectories without considering collaborative learning dynamics [21]. Moreover, studies have explored incorporating learning styles [22], recognizing that user preferences vary based on cognitive and behavioral traits. Some works have highlighted mutual recommendation processes, where both students and the system influence learning paths [23]. Despite these efforts, existing techniques often fail to account for temporal user engagement and collaborative learning patterns. Collaborative learning patterns refer to behavioral regularities that emerge across multiple learners, such as co-enrollment trends, shared course transitions, or common progression pathways. Many existing approaches model learning paths at the individual level and do not explicitly incorporate such cross-learner dynamics, thereby treating learner trajectories as largely independent [13].

A critical gap remains in integrating sequential recommendation techniques into the learning path generation. Existing models such as [24–26] often suggest single-step recommendations rather than structured, goal-driven learning paths. Our work addresses this by leveraging sequential recommender embeddings and projecting them into an LLM to generate dynamic and personalized learning trajectories.

Next, we delineate the works in the field of sequential recommendations and LLM-based techniques.

2.2. Sequential recommendation

Sequential recommendation systems model user–item interactions over time, predicting the next most relevant item based on past behavior. Early methods, such as association rule mining [27] and Markov chains [28], extracted frequent sequential patterns but were limited by rigid rule-based dependencies. To address these limitations, later works incorporated user and item metadata [29] and temporal dynamics [24], allowing models to predict future preferences more effectively. Deep learning further revolutionized the field, with RNN-based models [25,30] significantly outperforming traditional methods. However, RNN-based models struggled with long-term dependencies and vanishing gradients.

The introduction of Transformer-based models, such as SASRec [24] and BERT4Rec [26], addressed these issues by employing self-attention mechanisms. These models capture long-range dependencies in user behavior and contextual relationships between interactions. However, their reliance on fixed architectures limits their ability to leverage external knowledge sources. Additionally, convolutional models, such as Caser [31], introduced positional importance in sequences but remained constrained by their fixed structure.

A key research gap exists in the systematic fusion of sequential recommendation techniques with LLM-based models. Current methods either focus on interaction sequences or external knowledge utilization, but few combine both effectively. Our approach bridges this gap by using sequential recommender embeddings as inputs for an LLM, allowing for context-aware learning path recommendations that dynamically adapt to user progress.

2.3. LLMs for recommendation

Transformer-based architectures have led to significant improvements in sequential recommendation, but their fixed architectures and task-specific training paradigms limit their ability to adapt dynamically to new data. Models such as SASRec [24] and BERT4Rec [26] focus primarily on predicting the next item, while contrastive learning methods [32,33] attempt to enhance their robustness.

Recently, LLMs have emerged as powerful tools for recommendation tasks. Chat-Rec [34] utilizes ChatGPT's conversational abilities to facilitate recommendations, while studies like [35,36] employ in-context learning to enable LLMs to adapt to recommendation tasks. However, since LLMs are not inherently trained for recommendation, their performance remains suboptimal without fine-tuning or instruction tuning. Some works have explored instruction tuning on empirical recommendation datasets [18,37], improving LLMs' ability to generate structured recommendations. Others have fine-tuned LLMs [38,39] or designed prompting techniques to enable tool utilization. However, most approaches fail to integrate sequential patterns systematically, resulting in less adaptive and less personalized recommendations.

2.4. Key contributions and research gap

Despite substantial progress in course recommendation and learning path modeling, a clear methodological gap remains. Sequential recommendation models such as SASRec [24], BERT4Rec [26], GRU4Rec [25] have demonstrated strong performance in next-item or next-course prediction. However, these models are fundamentally designed for single-step recommendation and do not explicitly support the generation of structured, multi-step learning paths that reflect longer-term learner progression.

Parallel to this line of work, recent studies have explored the use of large language models for educational recommendation and learning assistance. Although LLM-based approaches excel at generating coherent textual suggestions, they typically operate independently of progression-aware sequential recommender systems and rely primarily on prompt-level textual context. Consequently, they lack explicit grounding in learners' historical interaction sequences and thus fail to leverage the rich behavioral representations learned by sequential recommendation models.

These limitations highlight an open research gap at the intersection of sequential recommendation and learning path generation: existing methods either model learner behavior without producing coherent learning trajectories, or generate textual learning paths without embedding sequential behavioral signals. Addressing this gap requires a unified framework that integrates progression-aware sequential representations with language-model-based generation mechanisms. In contrast, our approach aims to -

- bridge sequential recommendation with LLMs, allowing for dynamic learning path generation rather than isolated recommendations.

- utilize learned embeddings from a sequential model to enhance LLM-based reasoning, improving personalization and adaptivity.
- incorporate prior user engagement, ensuring that learning paths are not just based on similar users but also on temporal behavior patterns.

This work contributes to the development of a unified framework that harnesses the strengths of both sequential models and LLMs, setting a new standard for context-aware and personalized learning recommendations.

2.5. Scope and delineation of applicability

The proposed framework is designed specifically for MOOC-based online learning environments, where learners interact with digital audio-video course content and platform-recorded behavioral traces are available. The personalization mechanism relies on sequential interaction histories, user preferences, and course-level metadata, which are intrinsic to MOOC platforms.

Given these assumptions, the current framework is not directly applicable to hybrid or face-to-face classroom learning environments. In such contexts, learner interactions are often mediated through in-person activities, informal discussions, or instructor-led assessments, which are not captured with comparable granularity or consistency in platform logs. Consequently, key signals required by our model — such as detailed course interaction sequences and engagement behaviors — may be incomplete or unavailable.

We therefore position this work as a solution tailored to MOOC-based online learning scenarios. Extending the framework to hybrid or classroom-based settings would require additional data collection mechanisms and methodological adaptations, such as explicit modeling of instructor-defined curricula or in-class learning activities, which are beyond the scope of the present study.

3. Problem formulation

Let $\mathcal{U} = \{u_1, u_2, \dots, u_m\}$ be the set of users and $C = \{c_1, c_2, \dots, c_n\}$ be the set of available courses. Each user $u \in \mathcal{U}$ interacts with a chronological sequence of courses, represented as:

$$S_u = [c_1, c_2, \dots, c_t] \tag{1}$$

where S_u denotes the user's learning history up to time step t. The goal of a sequential recommender system is to predict the next course c_{t+1} that the user is likely to engage with, given their historical interactions:

$$c_{t+1} = f_{\text{Rec}}(S_u) \tag{2}$$

where f_{Rec} is the learned function of the sequential recommendation model.

However, instead of predicting just the next course, we aim to generate an entire learning path $\mathcal{P}_u$ that aligns with the user's learning objectives. Formally, given a user's interaction history S_u, our goal is to generate an optimal learning sequence:

$$\mathcal{P}_u = [c_{t+1}, c_{t+2}, \dots, c_{t+k}] \tag{3}$$

where k is the number of future recommendations. The learning path should be both coherent and personalized, considering course content, user preferences, and sequential learning dependencies. To achieve this, we integrate large language models (LLMs) into the recommendation process. Specifically, we first use a sequential recommender to predict embeddings of potential next courses and then leverage an LLM to generate a structured learning path by modeling long-term dependencies and knowledge representations.

4. Methodology and solution overview

This work adopts an empirical, model-driven methodology in which a learning path recommendation framework is designed and evaluated using real-world MOOC interaction data. The methodology combines sequential modeling, representation learning, and large language model conditioning, followed by experimental validation against established baselines using standard evaluation metrics.

The research design follows a pipeline-based architecture, beginning with behavioral pattern extraction from historical interaction sequences and progressing toward generative learning path construction. The methodology is structured to ensure both predictive accuracy (through sequential modeling) and contextual richness (through LLM-based generation). Furthermore, the framework is evaluated under a controlled experimental setting to ensure reproducibility, robustness, and fair comparison with state-of-the-art methods.

Our approach consists of two key stages:

4.1. Sequential recommendation for next-course prediction

We employ a sequential recommender system to capture user interaction patterns and predict the most relevant next course c_{t+1}. The recommender learns latent embeddings for courses and user interactions, encoding both content and sequential learning preferences. This stage serves as the behavioral foundation of the framework, ensuring that recommendations are grounded in empirical interaction data rather than solely relying on semantic similarity. The output of this component is a set of behavior-aware course embeddings that encode personalized learning signals.

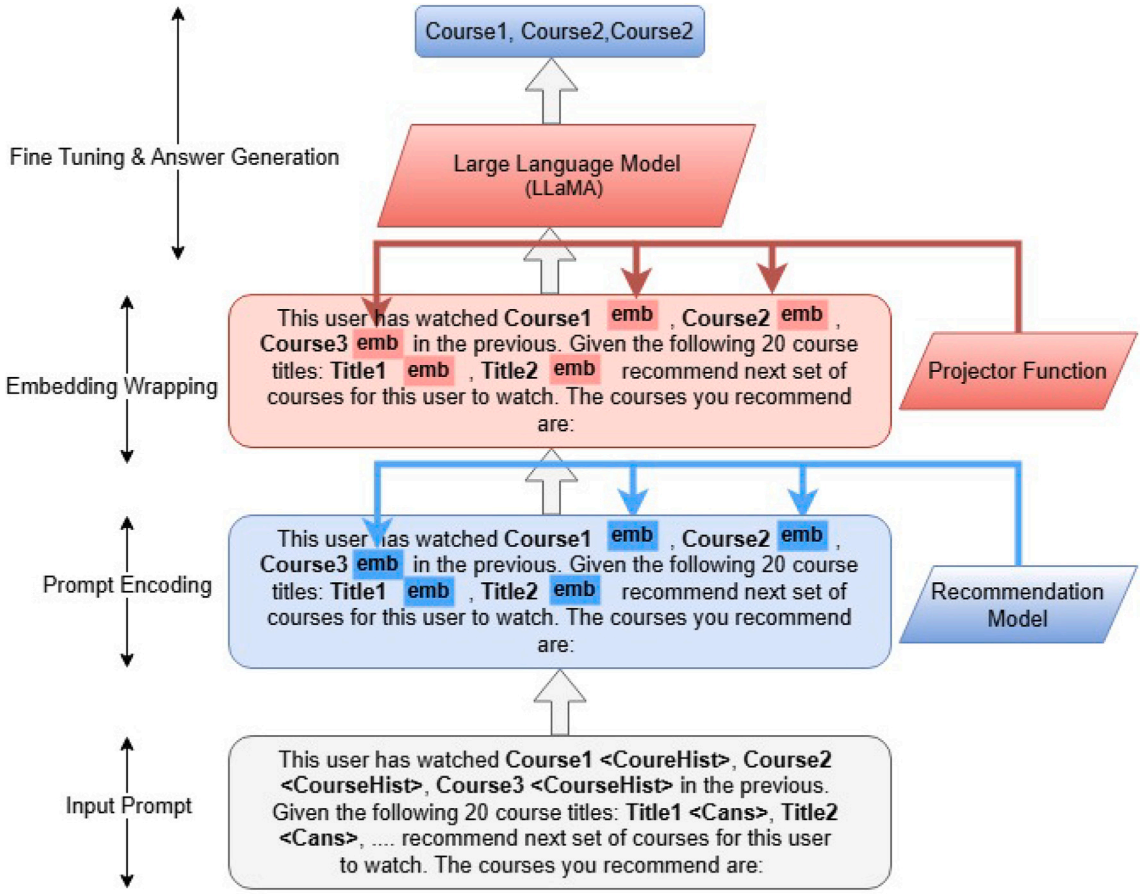


Fig. 1. Solution architecture.

4.2. LLM-based learning path generation

The learned course embeddings are projected into the LLaMA model to serve as contextual inputs. LLaMA utilizes its knowledge of course relationships and learning trajectories to generate an extended learning path $\mathcal{P}_u$ beyond just the next course. This allows for structured and goal-oriented recommendations, ensuring that the suggested learning path aligns with the user's past behavior and broader learning objectives. In this stage, the sequentially derived embeddings are transformed into a representation space compatible with the LLM, enabling conditioning of the generative process on behavior-aware signals. This integration allows the language model to incorporate both semantic understanding and personalized sequential patterns when constructing learning paths.

By combining sequential recommendations with LLM-based text generation, we move beyond simple next-item prediction to curating entire learning journeys that enhance user engagement and learning outcomes. The solution overview is depicted in Fig. 1.

5. Solution details

5.1. Sequential recommendation model

Herein, we present the sequential recommendation model named "Sequential Learning with User Insights" (SeqInsight). The proposed model, SeqInsight aims to derive comprehensive representations of both courses and user preferences by leveraging the textual content of user reviews and associated ratings. For each user u, the model infers the user's preferences by analyzing aspect-based sentiment scores extracted from their reviews, in conjunction with their rating patterns. These aspect scores reflect the user's evaluations of various course attributes, such as content quality, instructor effectiveness, and difficulty level, enabling a nuanced understanding of their preferences. Formally, the user preference representation $\mathbf{u}_{\text{pref}} \in \mathbb{R}^d$ is derived as:

$$\mathbf{u}_{\text{pref}} = f_{\text{pref}}(\{\mathbf{s}_{u,c} \mid c \in C_u\}), \tag{4}$$

where $\mathbf{s}_{u,c} \in \mathbb{R}^k$ denotes the aspect-based sentiment scores for user u and course c, C_u represents the set of courses reviewed by user u, and f_{pref} is a preference aggregation function (e.g., averaging or neural aggregation).

Similarly, each course c is represented by aggregating the aspect-based sentiment scores derived from all associated reviews. This process leverages a methodology inspired by the architecture proposed by Li et al. [40] for aspect-based sentiment extraction. The course representation $\mathbf{c} \in \mathbb{R}^d$ is computed as:

$$\mathbf{c} = f_{\text{course}}(\{\mathbf{s}_{u,c} \mid u \in \mathcal{U}_c\}), \tag{5}$$

where $\mathcal{U}_c$ denotes the set of users who reviewed course c, and f_{course} is an aggregation function that combines the aspect-based sentiment scores into a unified course embedding. By incorporating these implicit user perceptions, we complement explicit metadata (e.g., course title, category, and duration) to construct a more comprehensive and nuanced representation of the course. As illustrated in Fig. 2, the left branch of the architecture constructs static semantic representations for both users and courses, which are subsequently consumed by the sequential modeling component on the right. Specifically, aspect-based sentiment scores extracted from user reviews and ratings are first encoded via a CNN and projected through a trainable weight matrix to obtain a fixed-dimensional user preference representation $\mathbf{u}_{\text{pref}}$. In parallel, review-derived aspect scores associated with each course are processed through an analogous CNN-based pipeline to produce the corresponding course representation $\mathbf{c}$. These course representations are ordered according to each user's interaction sequence and serve as inputs to the multi-headed self-attention module, where they form the query, key, and value matrices to model temporal dependencies in the user's learning history. The attention outputs are concatenated and linearly transformed to yield the user learning history representation, which captures sequential behavioral patterns. Finally, this sequential representation is combined with the static user preference representation via weighted vector addition, and the resulting unified user embedding is passed to the prediction head for next-course recommendation. We note that the proposed framework does not explicitly infer a learner's knowledge mastery state; instead, it leverages sequential behavioral signals and pedagogically relevant metadata as proxies for learning progression.

With respect to course content, the model takes as input the textual information associated with each course (e.g., review text and corresponding ratings), which is treated as a proxy for course video content rather than directly processing raw video signals. Feature extraction is performed using a task-specific CNN encoder trained jointly with the recommendation objective, which learns discriminative representations from the review corpus and associated sentiment signals. As shown in Fig. 2, this CNN-based feature extractor replaces the need for external pre-trained models by directly learning domain-specific representations optimized for downstream recommendation. Consequently, all course representations are learned end-to-end within the proposed framework.

5.1.1. Multi-head self-attention over user history

To account for the dynamic nature of user preferences over time, the model incorporates a *multi-head self-attention mechanism* that operates over the sequence of courses recently completed by the user. This mechanism allows the model to prioritize and weigh the influence of recent interactions, capturing temporal shifts in user preferences and enhancing the personalization of recommendations.

Let $\mathcal{H}_u = \{c_1, c_2, \ldots, c_L\}$ denote the sequence of L courses recently completed by user u, where each course c_m is represented by its embedding $\mathbf{c}_m \in \mathbb{R}^d$. For a candidate course c, the attention weight α_m is calculated as:

$$\alpha_m = \frac{\exp(\mathbf{c}_m^\top \mathbf{c})}{\sum_{n=1}^{L} \exp(\mathbf{c}_n^\top \mathbf{c})}. \tag{6}$$

The historical learning pattern of user u with respect to candidate course c is computed as a weighted sum of the course embeddings:

$$\mathbf{u}_{\text{hist}} = \sum_{m=1}^{L} \alpha_m \mathbf{c}_m. \tag{7}$$

Although the final user representation is computed via a weighted aggregation, the weights are learned through a multi-head self-attention mechanism, enabling the model to capture non-linear, context-dependent relationships such as temporal preference shifts and latent course dependencies rather than a simple average.

5.1.2. User representation and preference scoring

The final user representation $\mathbf{u} \in \mathbb{R}^{2d}$ is obtained by concatenating the user's historical representation $\mathbf{u}_{\text{hist}}$ and their preference representation $\mathbf{u}_{\text{pref}}$:

$$\mathbf{u} = [\mathbf{u}_{\text{hist}}; \mathbf{u}_{\text{pref}}]. \tag{8}$$

This combined representation $\mathbf{u}$ is then used to compute the preference score $y_{u,c}$ of user u for a candidate course c using a sigmoid function:

$$y_{u,c} = \sigma(\mathbf{u}^\top \mathbf{c}), \tag{9}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function. The score $y_{u,c} \in [0, 1]$ reflects the likelihood of user u preferring course c, enabling personalized course recommendations.

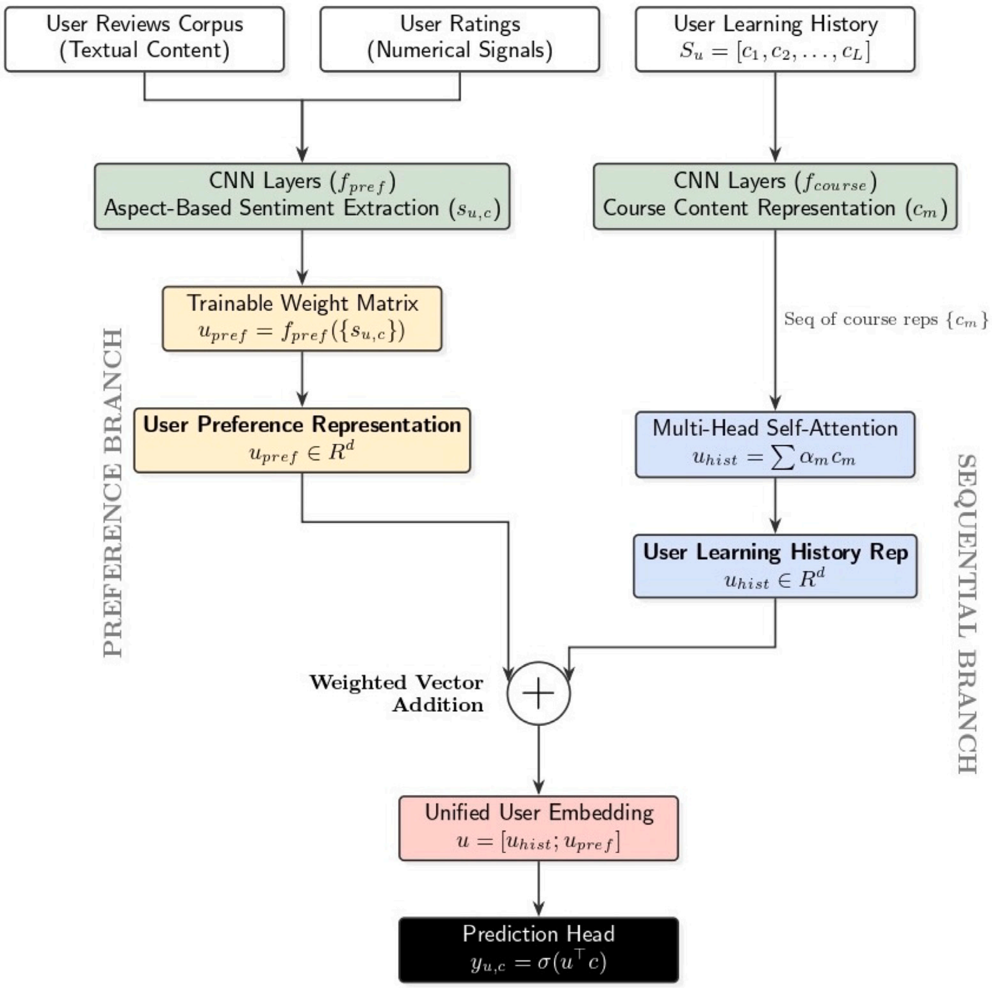


Fig. 2. Model architecture of SeqInsight.

5.1.3. Training objective

The training objective of the proposed model is to minimize the binary cross-entropy loss, which measures the discrepancy between the predicted preference scores and the actual user interactions. For each user u, we consider the sequence of courses they have interacted with, denoted as $\mathcal{H}_u = \{c_1, c_2, \ldots, c_L\}$, where L is the length of the sequence. The model is trained to predict the user's preference for the last course c_L in the sequence, given the preceding courses $\{c_1, c_2, \ldots, c_{L-1}\}$.

Formally, let y_{u,c_L} denote the predicted preference score of user u for the last course c_L, computed as:

$$y_{u,c_L} = \sigma(\mathbf{u}^\top \mathbf{c}_L), \tag{10}$$

where $\mathbf{u}$ is the final user representation derived from the sequence $\{c_1, c_2, \ldots, c_{L-1}\}$, and $\mathbf{c}_L$ is the embedding of the last course c_L. The ground truth label y_{u,c_L} is binary, where $y_{u,c_L} = 1$ if user u interacted with course c_L, and $y_{u,c_L} = 0$ otherwise.

The binary cross-entropy loss $\mathcal{L}$ for user u is defined as:

$$\mathcal{L}(u) = -\left(y_{u,c_L} \log(\hat{y}_{u,c_L}) + (1 - y_{u,c_L}) \log(1 - \hat{y}_{u,c_L}) \right). \tag{11}$$

To train the model, we minimize the average loss over all users and their respective sequences in the training set. The overall training objective is:

$$\mathcal{L} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \mathcal{L}(u), \tag{12}$$

where $\mathcal{U}$ is the set of all users in the training data. This objective encourages the model to accurately predict user preferences for the last item in each sequence, leveraging the historical interactions to capture evolving user preferences over time.

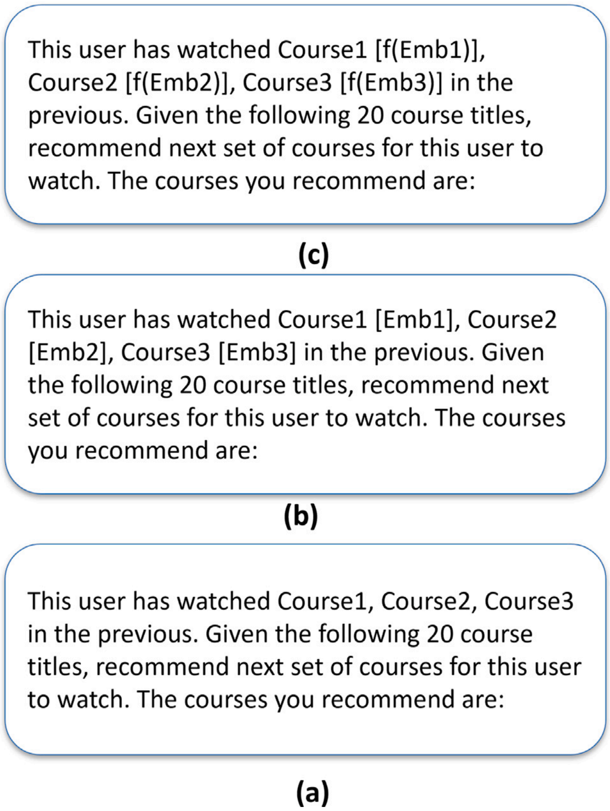


Fig. 3. Illustration of prompt construction and embedding alignment. (a) Raw text-based prompt constructed using course titles from the user's interaction history. (b) Prompt augmented with course embedding placeholders $[\mathrm{Emb}_i]$ corresponding to embeddings produced by the recommendation model. (c) Final prompt in which the original embeddings are transformed using the projection function f, resulting in projected embeddings $[f(\mathrm{Emb}_i)]$ that are aligned with the language model's input modality.

5.2. *Learning path generation model*

This section describes the second stage of our model, i.e., learning path generation using generative model. We present our model, "Personalized Learning Path Generation with LLM" (LearnGPT).

5.2.1. *Prompt construction*

To generate learning paths based on a user's watch history, we design a text-based prompt template. This template incorporates the titles of courses completed by the user to provide historical context. Additionally, to embed behavioral information, we include user and course embeddings learned from the recommendation model. However, since the language model's prompt and the user vector representation from the recommendation model reside in different modalities, it is crucial to ensure alignment between these modalities. This alignment allows the language model to interpret the prompt correctly and utilize the user's learning style to generate tailored recommendations.

To achieve this alignment, we introduce a projection function f. This function maps the user and item vectors into a shared modality compatible with the language model. Specifically, f is implemented as a two-layer shallow feedforward network that learns a trainable projection of these vectors. Mathematically, the projection function can be expressed as:

$$\mathbf{z}_u = f(\mathbf{u}), \quad \mathbf{z}_c = f(\mathbf{c}), \tag{13}$$

where $\mathbf{u}$ and $\mathbf{c}$ represent the user and course embeddings, respectively, and $\mathbf{z}_u$ and $\mathbf{z}_c$ are their projected counterparts in the language model's modality. The final prompt is constructed as depicted in Fig. 3.

As shown in Fig. 3, Fig. 3(a) illustrates the initial raw prompt constructed by inserting the course names from the user's interaction history into the prompt template. Fig. 3(b) presents the corresponding course embeddings produced by the recommendation model. Fig. 3(c) demonstrates the projected embeddings obtained by applying the projection function f to the original recommendation-model embeddings, thereby, aligning them with the modality required by the language model.

This approach ensures that the language model can effectively interpret and utilize the behavioral data to generate personalized learning paths.

5.2.2. Instruction tuning

In this work, we address the task of predicting the next item in a sequence based on a user's watch history by fine-tuning the LLaMA (Large Language Model Meta AI) model using instruction tuning. Instruction tuning is a methodology that adapts pre-trained language models to specific tasks by training them on datasets containing instruction-output pairs. The goal is to enable the model to understand and adapt to the learning style patterns of users, thereby improving its ability to generate accurate and contextually relevant predictions.

Given a sequence of items $x = (x_1, x_2, \dots, x_n)$ representing a user's watch history, the task is to predict the next item $y = (y_1, y_2, \dots, y_T)$ in the sequence. The model is trained to generate y conditioned on x, ensuring that the output sequence aligns with the user's preferences and observed behavior. To achieve this, we fine-tune the LLaMA model on a dataset $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$, where $x^{(i)}$ is the input sequence (user watch history) and $y^{(i)}$ is the corresponding target sequence (next item to predict).

The training process involves minimizing a loss function $\mathcal{L}(\theta)$, which measures the discrepancy between the model's predictions and the ground truth. Specifically, we use the cross-entropy loss, defined as:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \log P(y_t^{(i)} | x^{(i)}, y_{<t}^{(i)}; \theta), \tag{14}$$

where θ represents the parameters of the model, N is the total number of training examples, T is the length of the target sequence, $x^{(i)}$ is the input sequence for the ith example, $y^{(i)}$ is the target sequence for the ith example, $y_t^{(i)}$ is the tth token in the target sequence, $y_{<t}^{(i)}$ denotes the tokens generated by the model before time step t, and $P(y_t^{(i)} | x^{(i)}, y_{<t}^{(i)}; \theta)$ is the probability assigned by the model to the correct token $y_t^{(i)}$ given the input $x^{(i)}$ and the previously generated tokens $y_{<t}^{(i)}$. The cross-entropy loss quantifies the model's ability to predict the correct token at each time step, and minimizing this loss corresponds to maximizing the likelihood of the target sequence.

During training, the model also updates the weights of a projector, which is responsible for aligning the embeddings with the text prompt space. This ensures that the embeddings are properly mapped to the space of text prompts, enabling the model to generate more accurate and contextually relevant predictions. The updated loss function, incorporating the projector's role, is given by:

$$\mathcal{L}(\theta, \phi) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \log P(y_t^{(i)} | x^{(i)}, y_{<t}^{(i)}; \theta, \phi), \tag{15}$$

where ϕ represents the parameters of the projector. The projector ensures that the embeddings are properly aligned with the text prompt space, enhancing the model's ability to generate accurate predictions.

The training objective is to find the optimal model parameters θ^* and projector parameters ϕ^* that minimize the loss function $\mathcal{L}(\theta, \phi)$:

$$\theta^*, \phi^* = \arg\min_{\theta,\phi} \mathcal{L}(\theta, \phi). \tag{16}$$

This optimization is performed using gradient-based methods such as Adam. By minimizing the cross-entropy loss, the model learns to align its predictions with the ground truth, thereby improving its ability to generate accurate and relevant predictions for the next item in the sequence.

This instruction tuning framework, enhanced with the projector, enables the LLaMA model to effectively capture user-specific patterns and adapt to their learning style, making it particularly suitable for applications in recommendation systems. Future work may explore extensions to this approach, such as incorporating additional contextual information or leveraging multi-task learning to further enhance the model's capabilities.

5.2.3. LoRA (Low-Rank Adaptation)

LoRA (Low-Rank Adaptation) [41] is a parameter-efficient fine-tuning method that introduces two trainable low-rank matrices to update the weights of a pre-trained language model (PLM). Specifically, LoRA employs a down-projection matrix $W_{\text{down}} \in \mathbb{R}^{d\times r}$ and an up-projection matrix $W_{\text{up}} \in \mathbb{R}^{r\times k}$ in parallel with the query (Q), key (K), and value (V) matrices in the attention layer of a transformer. For a pre-trained weight matrix $W \in \mathbb{R}^{d\times k}$, LoRA updates W using a low-rank decomposition:

$$\Delta W = W_{\text{down}} W_{\text{up}}, \tag{17}$$

where $r \ll \{d, k\}$. During training, the weights of the PLM are frozen, and only the low-rank matrices W_{down} and W_{up} are fine-tuned. This approach significantly reduces the number of trainable parameters, making it computationally efficient. During inference, the LoRA weights are merged with the original weight matrix W without increasing the inference time. Additionally, a scaling factor $s = \frac{1}{r}$ is often applied to the LoRA module to stabilize training.

5.2.4. QLoRA (Quantized LoRA)

QLoRA [42] is an extension of LoRA that further enhances computational efficiency by quantizing the transformer model to 4-bit NormalFloat (NF4) precision. This quantization reduces the memory footprint and computational requirements, enabling fine-tuning of large language models (LLMs) on hardware with limited resources. QLoRA employs double quantization to minimize memory usage and a paged optimizer to handle memory spikes during training. The NF4 data type is theoretically optimal for normally distributed weights, ensuring minimal loss in precision during quantization. Although the pre-trained weights W are quantized

Table 1
Statistics of the datasets.

Dataset	Computer Science	Data Science	Miscellaneous
Unique users	4495	3773	1387
Avg sequence length	6.46	6.21	5.66
Avg # of reviews	345.83	390.33	87.29
Unique courses	84	60	90
Number of sequences	4495	3773	1387
Number of interactions	29,050	23,420	7856
Max sequence length	549	259	208
Min sequence length	4	4	4

from FP16 to NF4, the auxiliary weights of the LoRA matrices W_{down} and W_{up} remain in FP16, ensuring that the final weights are restored to FP16 after fine-tuning. This allows QLoRA to achieve high performance while maintaining computational efficiency.

We employ LoRA (Low-Rank Adaptation) and QLoRA for instruction tuning to achieve efficient fine-tuning of large language models (LLMs). LoRA introduces trainable low-rank matrices W_{down} and W_{up} while freezing pre-trained weights, reducing trainable parameters and computational costs. QLoRA enhances this by quantizing weights to 4-bit NormalFloat (NF4) precision, enabling fine-tuning on hardware with limited resources. In our implementation, we use $r = 64$, $\alpha = 16$, and NF4 quantization, ensuring high performance while minimizing memory and computational requirements. This combination makes LoRA and QLoRA ideal for adapting LLMs to specific tasks efficiently.

6. Experimental details

6.1. Dataset

This section presents the dataset used to evaluate the proposed model. The dataset is obtained from Coursera,[1] a leading online learning platform that offers courses across various domains, provided by universities and institutions worldwide. It comprises user-generated reviews and ratings, along with course metadata such as *course name, university, difficulty level, course rating, course URL, course description, skills, review details*, and *domain* information.

The raw review data used in this study was initially obtained from a publicly available Coursera reviews dataset hosted on Kaggle.[2] This dataset contains approximately 1.45 million user reviews and ratings associated with Coursera courses. The core variables used from this dataset include `name`, `institution`, `course_url`, `course_id`, `reviews`, `reviewers`, `date_reviews`, and `rating`. These attributes capture essential information about the course identity, reviewer, temporal context, and user feedback. In addition to the Kaggle-provided attributes, supplementary course-level metadata, specifically *difficulty level* and *skills*, were independently scraped from the official Coursera platform to enrich the dataset and support more informative course representations.

To facilitate the structured evaluation, the dataset is organized into three distinct domains including Computer Science, Data Science, and other subject areas, as summarized in Table 1. Each domain represents a broad category of courses, enabling the evaluation of the model across diverse subject areas. Since the task focuses on sequence generation, a user's review history serves as a proxy for their course interaction history. To ensure meaningful sequence generation, we impose a minimum sequence length of 4. For users with longer review histories, we employ a sliding window approach to generate additional sequences, ensuring adequate sequence length while minimizing overlap.

To ensure a robust evaluation, the dataset is split at the user level, ensuring that no overlapping sequences appear across the training, validation, and test sets. The dataset is partitioned in a 70:15:15 ratio into training, validation, and test sets. This partitioning strategy ensures that the model is evaluated on unseen user sequences, providing a reliable assessment of its generalization capabilities.

6.2. Experimental settings

All experiments were conducted on a Windows system with a 2.7 GHz Intel Xeon processor and 16 GB RAM with NVIDIA Tesla K80 GPU. Hyperparameters were optimized using a grid search algorithm, with the final values selected based on the best-performing configurations from preliminary experiments.

[1] https://www.coursera.org/.
[2] https://www.kaggle.com/datasets/imuhammad/course-reviews-on-coursera?select=Coursera_reviews.csv.

6.3. Evaluation metrics

6.3.1. Evaluation metrics for SeqInsight

The evaluation metrics used for next course prediction include Mean Reciprocal Rank (MRR), Hit@K, Normalized Discounted Cumulative Gain (NDCG@K), and Precision@K. These metrics collectively assess the model's ability to accurately predict the next item in a sequence by evaluating both ranking quality and recommendation relevance.

1. **Precision@K** measures the proportion of relevant items among the top-K recommendations:

$$\text{Precision@K} = \frac{\text{Number of relevant items in top-K}}{K} \tag{18}$$

2. **Mean Reciprocal Rank (MRR)** measures ranking quality based on the reciprocal rank of the first relevant item:

$$\text{MRR} = \frac{1}{|Q|}\sum_{i=1}^{|Q|}\frac{1}{\text{rank}_i} \tag{19}$$

where rank_i is the position of the first relevant item for the ith query, and $|Q|$ is the total number of queries.

3. **Hit@K** indicates whether the correct item appears in the top-K recommendations:

$$\text{Hit@K} = \begin{cases} 1, & \text{if the relevant item is in the top-K recommendations} \\ 0, & \text{otherwise} \end{cases}$$

4. **NDCG@K** evaluates ranking quality by assigning higher importance to relevant items appearing earlier in the top-K recommendations:

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}} \tag{20}$$

where:

$$\text{DCG@K} = \sum_{i=1}^{K}\frac{2^{\text{rel}_i}-1}{\log_2(i+1)}$$

and IDCG@K (Ideal DCG@K) is computed by sorting relevant items in descending order of relevance.

6.3.2. Evaluation metrics for LearnGPT

Traditional recommendation metrics like Recall, Precision, and MRR focus on single-item predictions and ranking, making them less effective for assessing sequential dependencies. Given that our task involves sequence generation, we adopt the ROUGE (Recall-Oriented Understudy for Gisting Evaluation) score as the primary evaluation metric. ROUGE is a widely used metric in natural language processing (NLP) for evaluating the quality of generated sequences by measuring n-gram, word sequence, and word pair overlap with reference sequences, making it well-suited for evaluating coherence and relevance. In our experiments, we use the most commonly used ROUGE variants including ROUGE-N and ROUGE-L. The following are the definitions.

1. ROUGE-N (N-gram Overlap) ROUGE-N measures the n-gram recall between a generated sequence and a reference sequence:

$$\text{ROUGE-N} = \frac{\sum_{s\in S}\sum_{w\in W_s}\min(\text{Count}_{\text{gen}}(w), \text{Count}_{\text{ref}}(w))}{\sum_{s\in S}\sum_{w\in W_s}\text{Count}_{\text{ref}}(w)} \tag{21}$$

where: S is the set of reference sequences. W_s is the set of n-grams in reference sequence s. $\text{Count}_{\text{gen}}(w)$ is the number of times n-gram w appears in the generated sequence, and $\text{Count}_{\text{ref}}(w)$ is the number of times w appears in the reference sequence. In our evaluation we used **ROUGE-1** and **ROUGE-2**.

2. ROUGE-L (Longest Common Subsequence - LCS) ROUGE-L evaluates sequence similarity based on the Longest Common Subsequence (LCS), which considers sentence-level structure without requiring exact word matches:

$$\text{ROUGE-L} = \frac{LCS(\text{gen}, \text{ref})}{|\text{ref}|} \tag{22}$$

where: $LCS(\text{gen}, \text{ref})$ is the length of the longest common subsequence between the generated sequence and the reference sequence and $|\text{ref}|$ is the length of the reference sequence.

6.4. Baseline approaches

The baseline models used in this study are discussed next.

Table 2
Performance comparison of different recommendation models across various metrics on Computer Science dataset.

	Model	MRR	NDCG	Hit	Precision
K = 5	GRU4Rec	0.167	0.200	0.302	0.060
	SASRec	0.191	0.255	0.454	0.091
	Bert4Rec	0.149	0.186	0.299	0.059
	LightSAN	0.187	0.252	0.452	0.090
	FEHARN	0.195	0.254	0.437	0.087
	SeqInsight	**0.391**	**0.279**	**0.679**	**0.203**
	Improvement %	100.51	9.41	49.56	123.08
K = 10	GRU4Rec	0.200	0.274	0.538	0.054
	SASRec	0.210	0.300	0.591	0.059
	Bert4Rec	0.170	0.238	0.458	0.046
	LightSAN	0.210	0.298	0.594	0.059
	FEHARN	0.210	0.300	0.579	0.058
	SeqInsight	**0.360**	**0.339**	**0.831**	**0.147**
	Improvement %	71.43	13	39.9	149.15
K = 15	GRU4Rec	0.203	0.298	0.629	0.042
	SASRec	0.216	0.322	0.677	0.045
	Bert4Rec	0.180	0.271	0.583	0.039
	LightSAN	0.213	0.320	0.680	0.045
	FEHARN	0.222	0.324	0.671	0.045
	SeqInsight	**0.335**	**0.366**	**0.881**	**0.115**
	Improvement %	50.9	12.96	29.56	155.56
K = 20	GRU4Rec	0.207	0.315	0.697	0.035
	SASRec	0.220	0.338	0.742	0.037
	Bert4Rec	0.184	0.286	0.645	0.032
	LightSAN	0.217	0.337	0.749	0.037
	FEHARN	0.225	0.339	0.735	0.037
	SeqInsight	**0.321**	**0.382**	**0.903**	**0.094**
	Improvement %	42.67	12.68	20.56	154.05

6.4.1. Baseline approaches for SeqInsight

1. GRU4Rec [25]: This seminal work pioneered deep learning for next-item prediction, using RNNs to model state transitions and Bayesian Personalized Ranking (BPR) as the loss function.
2. SASRec [24]: This work bridges Markov Chain-based models (MCs) and Recurrent Neural Networks (RNNs) for sequential recommendations by introducing SASRec, a self-attention-based model. Like RNNs, it captures long-term semantics by enabling multi-head self-attention mechanism for item transition.
3. Bert4Rec [26]: BERT4Rec is based on the Transformer architecture. Inspired by BERT, it models items as tokens and interaction sequences as sentences. Unlike traditional auto-regressive approaches, it leverages full bidirectional self-attention to capture patterns in user behavior.
4. LightSAN [43]: LightSANs is a sequential recommendation algorithm that speeds up the attention mechanism by projecting user history into k latent interests, replacing quadratic item-item interactions with linear item-interest attention for efficiency and robustness.
5. FEHARN [44]: FEARec unifies the time- and frequency-domain attention, using autocorrelation to detect periodic user behaviors and a frequency ramp to diversify pattern extraction, improving robustness and accuracy in sequential recommendations.

6.4.2. Baseline approaches for LearnGPT

To evaluate the effectiveness of our proposed model LearnGPT , we conduct comparisons against two widely recognized open-source language models: **LLaMA-2** Touvron et al. [45] (Meta, 2023) and **GPT-2** Radford et al. [46] (OpenAI, 2019). These models serve as appropriate benchmarks given their extensive use in the research community, transparent availability, and demonstrated competence across various natural language processing tasks.

1. LLaMA2 [45]: LLaMA-2 represents a significant evolution in publicly available language models, featuring a decoder-only transformer architecture with parameter counts spanning from 7 billion to 70 billion. Building on LLaMA-1, this iteration incorporates refined training datasets, increased context retention, and optimization through reinforcement learning from human feedback (RLHF).
2. GPT2 [46]: As one of the first large-scale autoregressive language models, GPT-2 with its 1.5 billion parameters played a pivotal role in the development of unsupervised pre-training techniques.

Table 3
Performance comparison of different recommendation models across various metrics on Data Science dataset.

	Model	MRR	NDCG	Hit	Precision
K = 5	GRU4Rec	0.113	0.142	0.232	0.046
	SASRec	0.168	0.219	0.378	0.076
	Bert4Rec	0.130	0.160	0.252	0.050
	LightSAN	0.152	0.202	0.358	0.072
	FEHARN	0.158	0.211	0.372	0.074
	SeqInsight	**0.398**	**0.354**	**0.730**	**0.272**
	Improvement %	136.9	61.64	93.12	257.89
K = 10	GRU4Rec	0.144	0.215	0.451	0.045
	SASRec	0.195	0.287	0.587	0.059
	Bert4Rec	0.154	0.221	0.445	0.044
	LightSAN	0.180	0.272	0.573	0.057
	FEHARN	0.185	0.275	0.569	0.057
	SeqInsight	**0.368**	**0.412**	**0.859**	**0.182**
	Improvement %	88.72	43.55	46.34	208.47
K = 15	GRU4Rec	0.152	0.243	0.558	0.037
	SASRec	0.204	0.314	0.689	0.046
	Bert4Rec	0.164	0.255	0.575	0.038
	LightSAN	0.188	0.298	0.672	0.045
	FEHARN	0.193	0.302	0.672	0.045
	SeqInsight	**0.341**	**0.445**	**0.920**	**0.142**
	Improvement %	67.16	41.72	33.5	208.7
K = 20	GRU4Rec	0.159	0.270	0.673	0.034
	SASRec	0.208	0.331	0.762	0.038
	Bert4Rec	0.170	0.280	0.681	0.034
	LightSAN	0.193	0.320	0.763	0.038
	FEHARN	0.198	0.323	0.761	0.038
	SeqInsight	**0.319**	**0.469**	**0.948**	**0.119**
	Improvement %	53.37	41.69	24.25	213.16

7. Experimental findings

7.1. Performance evaluation for SeqInsight

We evaluate the performance of our sequential recommendation system SeqInsight against five baseline models. The primary task of sequential recommendation is to predict the next item in a sequence given a user's historical interactions. To assess the effectiveness of our model, we compare it with state-of-the-art baselines using standard evaluation metrics discussed in Section 6.3.

We conducted a comprehensive evaluation of our model's performance against five state-of-the-art baselines across three datasets. The comparative analysis focused on four key metrics: Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG), Hit Rate, and Precision, evaluated for different values of K (K = 5, 10, 15, 20).

Our experimental results demonstrate consistent and substantial performance improvements across all datasets. Notably, for Computer Science dataset, the proposed model achieves particularly noteworthy gains as shown in Table 2. For K = 5, our model outperforms FEHARN (the strongest baseline for this metric) by 100% improvement in MRR. Additionally, SeqInsight also outperforms SASRec with relative improvements of 9.41% in NDCG@5, 49.56% in Hit@5, and 123.08% in Precision@5. The performance advantage persists across different recommendation lengths. For K = 10, where SASRec emerges as the strongest baseline (MRR = 0.210, NDCG = 0.300, Hit = 0.591, Precision = 0.059), SeqInsight maintains superior performance. Similar trends have been observed for K = 15 and K = 20 where our proposed model SeqInsight achieves an improvement of 50.9%, 42.67% in MRR, 12.96%, 12.68% in NDCG, 29.56%, 20.56% in Hit Rate, and 155.56%, 154.05% in Precision for K = 15 and K = 20, respectively.

The performance comparisons against the 'Data Science dataset' and the 'Miscellaneous Dataset' have been presented in Tables 3 and 4, respectively. As can be seen from the two tables SeqInsight achieves an improvement of up to 146.33% in MRR@5, 61.64% in NDCG@5, 125% in Hit@5, and 257.89% in Precision@5 across both the datasets.

The consistent performance advantage of our model stems from its innovative integration of review rating information with sequential pattern learning, addressing a critical limitation in existing approaches. While baseline models like *SASRec*, *LightSAN*, and *FEHARN* employ sophisticated attention mechanisms, they fundamentally overlook the valuable signal contained in user reviews. Our solution uniquely incorporates this rich feedback data to enhance personalization and recommendation accuracy. By simultaneously modeling both the temporal sequence of user interactions (*when* users engage) and the sentiment expressed in reviews (*how* they engage), the framework develops a more comprehensive understanding of user preferences.

To assess the performance contributions of the key model components, we conduct an ablation study, the results of which are reported in Appendix A.

Table 4
Performance comparison of different recommendation models across various metrics on Miscellaneous dataset.

	Model	MRR	NDCG	Hit	Precision
K = 5	GRU4Rec	0.177	0.214	0.324	0.065
	SASRec	0.176	0.208	0.301	0.060
	Bert4Rec	0.166	0.198	0.297	0.059
	LightSAN	0.175	0.207	0.306	0.061
	FEHARN	0.167	0.199	0.296	0.059
	SeqInsight	**0.436**	**0.294**	**0.729**	**0.205**
	Improvement %	146.33	37.38	125	215.38
K = 10	GRU4Rec	0.187	0.237	0.393	0.039
	SASRec	0.199	0.264	0.479	0.048
	Bert4Rec	0.183	0.242	0.433	0.043
	LightSAN	0.199	0.264	0.479	0.048
	FEHARN	0.188	0.252	0.464	0.046
	SeqInsight	**0.413**	**0.345**	**0.870**	**0.141**
	Improvement %	107.54	30.68	81.63	193.75
K = 15	GRU4Rec	0.191	0.249	0.441	0.029
	SASRec	0.206	0.289	0.572	0.038
	Bert4Rec	0.190	0.263	0.514	0.034
	LightSAN	0.206	0.290	0.578	0.039
	FEHARN	0.196	0.279	0.562	0.038
	SeqInsight	**0.384**	**0.371**	**0.932**	**0.114**
	Improvement %	86.41	27.93	61.25	192.31
K = 20	GRU4Rec	0.194	0.263	0.499	0.025
	SASRec	0.210	0.305	0.642	0.032
	Bert4Rec	0.195	0.286	0.611	0.031
	LightSAN	0.210	0.305	0.642	0.032
	FEHARN	0.200	0.296	0.635	0.032
	SeqInsight	**0.359**	**0.391**	**0.955**	**0.097**
	Improvement %	70.95	28.2	48.75	203.13

7.2. Performance evaluation for LearnGPT

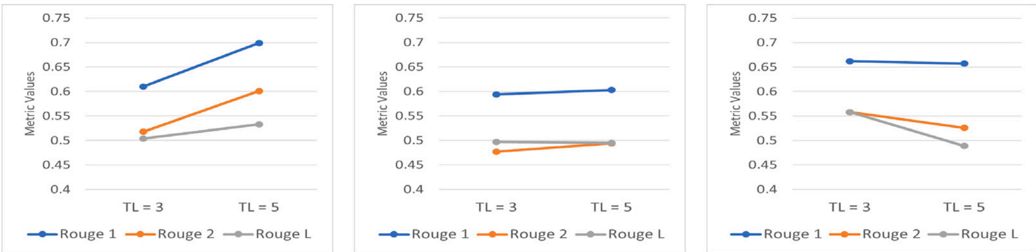
All computational experiments were performed on a Windows-based workstation equipped with an NVIDIA L4 Tensor Core GPU featuring 24 GB of GDDR6 memory. We employed a systematic grid search approach to optimize model hyperparameters, with the final configurations determined through extensive preliminary testing to identify the most effective settings. The selected values represent the optimal performance observed during these validation experiments. For the task of sequence generation, we employ *LLaMA-2* (Large Language Model Meta AI), an open-source large language model widely recognized for its robust performance and extensive use in academic research. LLaMA-2 is particularly suitable for our task due to its open-source nature, comprehensive documentation, and strong performance on a variety of natural language processing tasks.

The model is initialized with a learning rate of 0.0003, which was determined through preliminary experiments to balance convergence speed and stability. During training, we monitor the validation score after every 10 iterations to ensure the model is learning effectively and to avoid overfitting. We experimented with different numbers of training epochs, specifically 1, 3, and 5, to identify the optimal training duration. Due to memory constraints, the batch size is set to 1, as the data loader consumes significant memory during sequence generation tasks. This configuration ensures efficient utilization of computational resources while maintaining model performance.

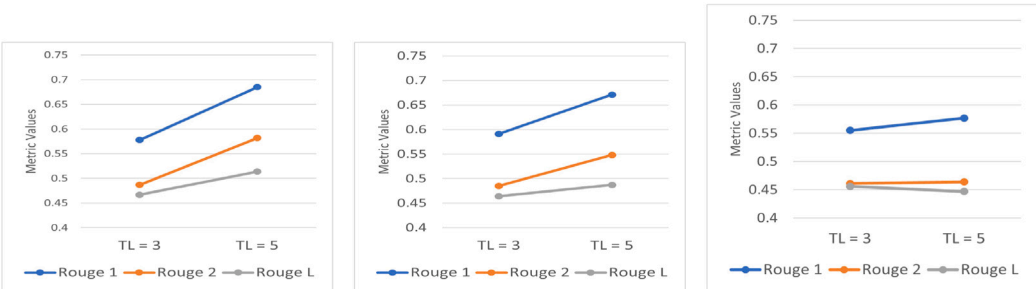
Our model demonstrates superior performance across all domains and metrics, achieving substantial improvements over both baseline models as shown in Table 5. In the Computer Science dataset, we observe particularly strong results with ROUGE-1 and ROUGE-2 scores of 0.610 and 0.518 respectively, representing improvements of 824.24% and 10 260% over GPT-2. The model maintains this advantage against LLaMA-2, outperforming it by significant margins (351.85% and 1626.67% for ROUGE-1 and ROUGE-2, respectively). The Data Science domain reveals similar trends, where our model achieves ROUGE-L score of 0.497, marking an 428.72% improvement over GPT-2 and 126.94% over LLaMA-2. Notably, the improvement percentages remain consistently high across all metrics, with the smallest being 425.66% for ROUGE-1 against GPT-2 in this domain. For the Miscellaneous dataset a similar trend has been observed. The consistent performance across diverse domains suggests our model's robust generalization capabilities. However, to empirically evaluate the generalization capability of the proposed framework across learning domains, we conduct another set of cross-domain transfer experiments, with results reported in Appendix B. Also to examine the impact of embedding-based course representations in LearnGPT, we perform a dedicated ablation study comparing models with and without course embeddings, as detailed in Appendix C.

Table 5
Comparison of ROUGE scores across datasets and models, showing absolute scores and percentage improvements over baseline models.

Dataset	Model	Rouge 1	Rouge 2	Rouge L
	GPT2	0.066	0.005	0.057
	Improvement %	824.24	10 260.00	784.21
Computer Science	Llama 2	0.135	0.030	0.116
	Improvement %	351.85	1626.67	334.48
	LearnGPT	**0.610**	**0.518**	**0.504**
	GPT2	0.113	0.038	0.094
	Improvement %	425.66	1155.26	428.72
Data Science	Llama 2	0.255	0.134	0.219
	Improvement %	132.94	255.97	126.94
	LearnGPT	**0.594**	**0.477**	**0.497**
	GPT2	0.123	0.052	0.109
	Improvement %	438.21	973.08	411.93
Miscellaneous	Llama 2	0.182	0.085	0.170
	Improvement %	263.74	556.47	228.24
	LearnGPT	**0.662**	**0.558**	**0.558**



(a) Computer Science (b) Data Science (c) Miscellaneous



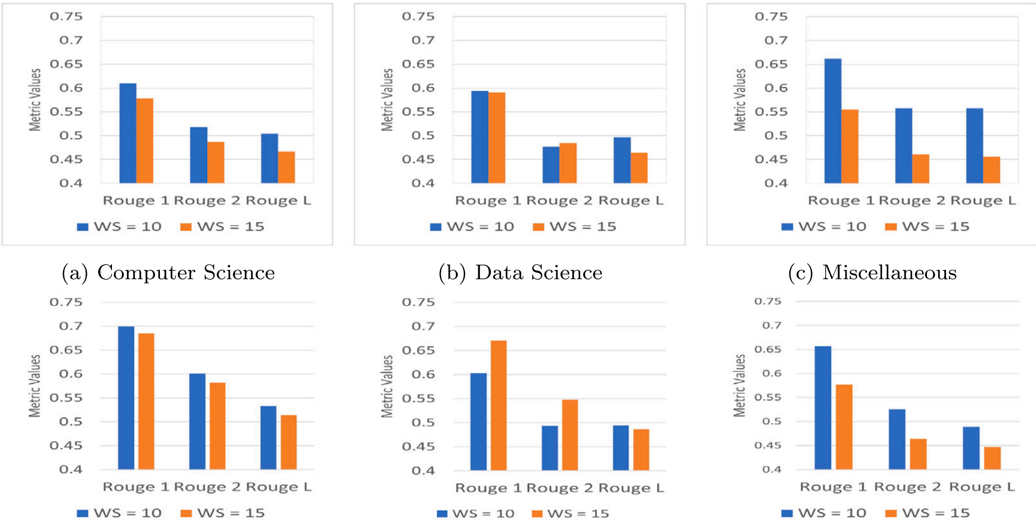
(d) Computer Science (e) Data Science (f) Miscellaneous

Fig. 4. Impact of window size on model performance across different datasets, (a)–(c) are for window size 10, (d)–(f) are for window size 15.

8. Discussion

8.1. Impact of window size

We have varied the target length to observe the impact of window size in the model performance. With a fixed target length of 3, model performance varies significantly across different window sizes. As can be seen from Fig. 4, for the Computer Science dataset, increasing the window size from 10 to 15 leads to a decline in all ROUGE metrics, with ROUGE-1 decreasing from 0.61 to 0.578 and ROUGE-L dropping from 0.504 to 0.467. A similar trend is observed in the Miscellaneous dataset, where ROUGE-1 falls from 0.662 to 0.555, and ROUGE-L decreases from 0.558 to 0.456, suggesting that larger windows may introduce noise or irrelevant context. However, the Data Science dataset remains relatively stable, with ROUGE-1 showing only a slight decrease from 0.594 to 0.591, and ROUGE-L reducing from 0.497 to 0.464. These results indicate that while increasing the window size theoretically provides more context, it can also dilute relevant information, leading to performance degradation in most domains—with Data Science being a key exception.



(a) Computer Science (b) Data Science (c) Miscellaneous

(d) Computer Science (e) Data Science (f) Miscellaneous

Fig. 5. Impact of target length on model performance across different datasets, (a)–(c) are for target length 3, (d)–(f) are for target length 5.

8.2. Impact of target length

We have also experimented the impact of target length on the performance of the model as shown in Fig. 5. As can be seen from the table, for a window size of 10, increasing the target length from 3 to 5 produces mixed results. The Computer Science dataset shows improved ROUGE-1 (increases from 0.61 to 0.699) but decreased ROUGE-L (from 0.504 to 0.533), suggesting longer predictions capture more content but with reduced coherence. The Data Science dataset exhibits similar gains in ROUGE-1 (from 0.594 to 0.603) while maintaining stable ROUGE-L (from 0.497 to 0.495). Conversely, the Miscellaneous dataset shows consistent reduction across all metrics. This pattern intensifies with window size 15, where Computer Science maintains its positive trajectory (ROUGE-1: $0.578 \rightarrow 0.685$) while Data Science shows even greater improvements (ROUGE-1: $0.591 \rightarrow 0.671$). For the Miscellaneous dataset ROUGE-L drops from 0.456 to 0.447. These findings suggest that target length expansion benefits structured domains (Computer Science and Data Science) but does not benefit diverse domains (Miscellaneous). The consistent improvement in Computer Science scores across both window sizes indicates this domain particularly benefits from longer-term dependencies in the prediction task.

8.3. Implications for research

This study highlights the potential of integrating sequential recommendation models with large language models (LLMs) for personalized learning path generation. Our approach bridges the gap between predictive recommendation techniques and generative AI, offering a novel direction for future research. First, it opens avenues for exploring hybrid models that dynamically adapt to user interactions over time, rather than relying on static course recommendations. Second, it underscores the importance of embedding transferability, where learned representations from sequential models can enhance LLM-based reasoning and decision-making. Third, our findings suggest that incorporating user engagement history into recommendation systems can significantly improve learning path relevance, emphasizing the need for more sophisticated temporal modeling techniques. Future research can build upon these insights by exploring alternative embedding techniques, fine-tuning LLMs with domain-specific knowledge, and designing explainable AI-driven learning path recommendations.

8.4. Implications for practice

From a practical perspective, our framework provides an adaptive solution for online education platforms, addressing key challenges in personalized learning. By leveraging sequential recommendation models, e-learning systems can offer dynamic learning paths that evolve with learners' progress, reducing cognitive overload and enhancing engagement. Additionally, integrating LLMs enables more nuanced and context-aware recommendations, allowing for more effective curriculum design and learner support. Educational institutions and MOOC platforms can adopt our approach to enhance course sequencing, optimize student learning journeys, and provide automated tutoring assistance. Moreover, the ability to capture both past behavior and generative reasoning makes this framework particularly useful for professional training programs, where learners follow non-traditional and flexible learning trajectories.

8.5. Implications for knowledge-based systems engineering

Beyond predictive performance, the proposed framework has implications for knowledge-based systems in educational engineering. The integration of sequential behavioral modeling with language-driven path synthesis demonstrates how heterogeneous AI components can be unified within a scalable systems architecture. In addition, embedding-level incorporation of collaborative learning signals enables the capture of community knowledge without explicit social network structures. From a systems engineering perspective, these properties contribute to adaptive, knowledge-centric learning infrastructures that align learner behavior and automated guidance. Overall, the framework advances the goal of knowledge-based systems engineering by enabling intelligent systems that transform distributed learning interactions into structured, actionable knowledge.

9. Conclusions and future work

In this work, we address an important gap in the current literature by bridging sequential course recommendation and learning path generation. Unlike prior sequential recommenders that focus on single-step prediction, our approach leverages embeddings learned from a progression-aware sequential model and projects them into a large language model to generate structured, multi-step learning paths. This integration enables the generation of personalized learning trajectories that are grounded in learners' historical interaction patterns rather than isolated course preferences. Overall, this work contributes to knowledge-based systems engineering by demonstrating how sequential learning analytics and generative language modeling can be integrated into a scalable architecture for adaptive learning path design.

Experimental results demonstrate significant improvements over traditional sequential models and state-of-the-art LLM-based approaches (with a relative improvements of up to 351.85% in ROUGE-1, 1626.67% in ROUGE-2, and 334.48% in ROUGE-L.), highlighting the effectiveness of our method in enhancing learning personalization.

Despite these promising results, several challenges remain. First, while our model effectively generates learning paths, further research is needed to enhance its interpretability and explainability for learners and educators. Second, expanding the model to incorporate multimodal data — such as video lectures, discussion forums, and assessments — could provide a more holistic understanding of learners' needs. Future work will explore these directions, aiming to refine and expand the proposed framework to support more diverse and complex learning environments. An important direction for future work is the integration of explicit knowledge-state modeling or pedagogical constraints (e.g., prerequisite graphs or difficulty level alignment) into the proposed framework. Furthermore, the proposed framework is evaluated only on real MOOC interaction data; while effective with limited user populations due to its emphasis on intra-user sequential dynamics, the potential benefits of synthetic data for extreme sparsity or cold-start scenarios are not explored and remain an avenue for future work.

CRediT authorship contribution statement

Prasad Deshmukh: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Harshit Samrat:** Software, Investigation, Formal analysis. **Nargis Pervin:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Ablation study

This appendix presents an ablation study to quantify the contribution of major architectural components in the proposed SeqInsight model. Ablated variants are created by removing one component at a time while keeping all other settings unchanged. Experiments are conducted on the Data Science dataset and evaluated using standard top-K recommendation metrics.

Tables A.1–A.4 report the results in Precision@K, MRR@K, NDCG@K, and Hit Rate@K. Removing the multi-head self-attention (MHA) mechanism leads to consistent and substantial performance degradation across all metrics. For example, Precision@5 drops from 0.2720 to 0.2617, NDCG@5 decreases from 0.3542 to 0.3332, and HR@5 declines from 0.7334 to 0.7024. This demonstrates the importance of attention in capturing sequential dependencies in learners' interaction histories.

Removing residual connections (RC) results in smaller but systematic performance losses. Specifically, Precision@5 decreases from 0.2720 to 0.2698, NDCG@5 from 0.3542 to 0.3502, and HR@5 from 0.7334 to 0.7236, indicating that residual connections primarily contribute to stabilizing representation learning rather than introducing new discriminative signals.

Table A.1
Ablation study reporting Precision@K on Data Science dataset.

Model	P@5	P@10	P@15	P@20
SeqInsight (Full)	**0.2720**	**0.1820**	**0.1426**	**0.1190**
w/o MHA	0.2617	0.1735	0.1376	0.1174
w/o RC	0.2698	0.1810	0.1409	0.1189
w/o MHA and RC	0.2671	0.1789	0.1400	0.1171

Table A.2
Ablation study: MRR@K results on Data Science dataset.

Model	MRR@5	MRR@10	MRR@15	MRR@20
SeqInsight	**0.3984**	**0.3680**	**0.3413**	**0.3190**
w/o MHA	0.3827	0.3565	0.3371	0.3137
w/o RC	0.3894	0.3697	0.3321	0.3248
w/o MHA and RC	0.3857	0.3605	0.3327	0.3243

Table A.3
Ablation study: NDCG@K results on Data Science dataset.

Model	NDCG@5	NDCG@10	NDCG@15	NDCG@20
SeqInsight (Full)	**0.3542**	**0.4121**	**0.4453**	**0.4690**
w/o MHA	0.3332	0.3869	0.4184	0.4461
w/o RC	0.3502	0.4090	0.4416	0.4622
w/o MHA and RC	0.3390	0.4005	0.4346	0.4570

Table A.4
Ablation study: Hit Rate@K results on Data Science dataset.

Model	HR@5	HR@10	HR@15	HR@20
SeqInsight (Full)	**0.7334**	**0.8590**	**0.9204**	**0.9483**
w/o MHA	0.7024	0.8329	0.9035	0.9457
w/o RC	0.7236	0.8576	0.9134	0.9472
w/o MHA and RC	0.7133	0.8451	0.9063	0.9416

Appendix B. Cross-domain generalization

To assess the generalization capability and transferability of LearnGPT across different educational domains, we conduct cross-domain evaluation experiments in which models fine-tuned on one domain are directly tested on datasets from other domains without additional fine-tuning. This setting reflects realistic deployment scenarios where labeled data may be unavailable for new domains.

Table B.1 report ROUGE-1, ROUGE-2, and ROUGE-L scores for all train–test domain combinations. As expected, in-domain evaluation (diagonal entries) yields the highest performance, with ROUGE-1 scores ranging from 0.594 to 0.662. Nevertheless, LearnGPT demonstrates meaningful cross-domain transfer, particularly between conceptually related domains. We quantify cross-domain transfer efficiency using a retention metric, defined as the percentage of in-domain ROUGE-1 performance preserved when a model fine-tuned on one domain is evaluated on another domain.

$$\text{Retention}(S \to T) = \frac{\text{ROUGE-1}(S \to T)}{\text{ROUGE-1}(S \to S)} \times 100\%.$$

Notably, strong bidirectional transfer is observed between the Computer Science (CS) and Data Science (DS) domains. A model fine-tuned on CS achieves a ROUGE-1 score of 0.405 when evaluated on DS, retaining 66.4% of its in-domain performance (0.610). Conversely, a DS-trained model achieves a ROUGE-1 score of 0.385 on CS data, retaining 64.8% of its in-domain performance (0.594). This mutual transferability aligns with the substantial conceptual overlap between these domains, including shared programming, algorithmic, and analytical foundations.

In contrast, transfer involving the Miscellaneous domain (MSC) exhibits greater asymmetry. While a MSC-trained model generalizes reasonably well to DS (ROUGE-1 = 0.424, retaining 64.0% of in-domain performance), the reverse transfer from DS to MSC is considerably weaker, achieving a ROUGE-1 score of 0.193 (32.5% retention). This discrepancy suggests that representations learned from technically focused curricula may not adequately capture the thematic and pedagogical characteristics of business-oriented learning paths, whereas broader-domain training may encode more transferable patterns.

Table B.2 summarizes cross-domain transfer efficiency in terms of retained ROUGE-1 performance. On average, LearnGPT retains 55.1% of in-domain performance across cross-domain settings, indicating that the model captures both domain-agnostic regularities and domain-specific characteristics.

These findings suggest that LearnGPT can generalize effectively across related educational domains without additional fine-tuning, supporting its practical applicability in scenarios where domain-specific training data are limited. However, for domains with substantial conceptual divergence, domain-specific fine-tuning remains important to achieve optimal performance.

Table B.1
Cross-domain ROUGE scores for LearnGPT. Rows indicate the fine-tuning domain and ROUGE metric, while columns denote the evaluation domain. Diagonal entries (in bold) represent in-domain performance; off-diagonal entries indicate cross-domain generalization without additional fine-tuning.

Train domain	Metric	CS	DS	MSC
CS	ROUGE-1	**0.610**	0.405	0.323
	ROUGE-2	**0.518**	0.303	0.255
	ROUGE-L	**0.504**	0.343	0.283
DS	ROUGE-1	0.385	**0.594**	0.193
	ROUGE-2	0.298	**0.477**	0.124
	ROUGE-L	0.343	**0.497**	0.177
MSC	ROUGE-1	0.329	0.424	**0.662**
	ROUGE-2	0.225	0.307	**0.558**
	ROUGE-L	0.286	0.359	**0.558**

Table B.2
Cross-domain transfer efficiency of LearnGPT measured as the percentage of in-domain performance retained when transferring from a source domain to a target domain. Higher percentages indicate stronger generalization capability.

Transfer direction	ROUGE-1 (%)	ROUGE-2 (%)	ROUGE-L (%)
CS → DS	66.4	58.5	68.1
CS → MSC	53.0	49.2	56.2
DS → CS	64.8	62.5	69.0
DS → MSC	32.5	26.0	35.6
MSC → CS	49.7	40.3	51.3
MSC → DS	64.0	55.0	64.3
Average	**55.1**	**48.6**	**57.4**

Table C.1
Ablation study comparing LearnGPT with and without course embeddings across domains.

Dataset	Model variant	ROUGE-1	ROUGE-2	ROUGE-L
Computer Science	With embeddings	**0.610**	**0.518**	**0.504**
	Without embeddings	0.462	0.358	0.386
Data Science	With embeddings	**0.594**	**0.477**	**0.497**
	Without embeddings	0.471	0.377	0.380
Miscellaneous	With embeddings	**0.662**	**0.558**	**0.558**
	Without embeddings	0.482	0.381	0.384

Appendix C. Effectiveness of course embeddings in LearnGPT

To empirically assess the contribution of course embedding representations in LearnGPT, we conduct an ablation study comparing models trained with and without course embeddings. In the embedding-based variant, LearnGPT incorporates dense course embeddings learned by the SeqInsight recommendation model and projected into the language model's input space via a trainable projection network. In contrast, the ablated variant relies solely on textual course titles using placeholder tokens, without embedding augmentation. Both variants are trained and evaluated under identical experimental settings.

Across all three domains, the embedding-based LearnGPT consistently outperforms against non-embedding counterpart. On the Computer Science dataset, ROUGE-1, ROUGE-2, and ROUGE-L improve from 0.462, 0.358, and 0.386 to 0.610, 0.518, and 0.504, respectively (Table C.1) Similar gains are observed on the Data Science and Miscellaneous, with average improvements of 31.7% in ROUGE-1, 36.5% in ROUGE-2, and 38.1% in ROUGE-L.

These results demonstrate that course embeddings provide benefits beyond surface-level textual cues. By encoding collaborative and sequential learning patterns derived from learner interaction histories, the embeddings capture semantic relationships among courses that are not fully expressed in course titles alone. Moreover, the use of dense embedding representations reduces sparsity associated with large course catalogs, thereby improving generalization.

Data availability

Data will be made available on request.

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